

mzCache: On-Device LLM Memory Management under Multitasking

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ABSTRACT

On-device mobile Large Language Model (LLM) inference is gaining significant attention. However, mobile devices operate in highly dynamic multitasking environments where users frequently switch between applications. This creates memory pressure, forcing LLM memory—model weights and KV cache—to be evicted by the operating system. When a new inference request arrives, the inference system must restore the evicted memory through slow storage reads or recompute the entire KV cache, severely degrading responsiveness. To address this, we present *mzCache*, an LLM inference system with specialized memory management for responsive inference in mobile multitasking environments. Under unpredictable memory pressure, *mzCache* elastically evicts LLM memory and leverages the unified memory of mobile SoCs to enable *zero-wait* inference on the GPU with concurrent CPU-side restoration. *mzCache* realizes this through restoration-oriented memory management: LLM memory is partitioned into fine-grained shared buffers to enable partial eviction and restoration with concurrent cross-processor access, while hybrid swap and backward-out eviction policies ensure low-latency restoration from any eviction state. Implemented on llama.cpp and deployed as an Android application, *mzCache* achieves 2.1–5.5× reduction in Time-to-First-Token compared to storage-backed partial offload and demonstrates its effectiveness in real multitasking scenarios.

CCS Concepts

• **Human-centered computing** → Ubiquitous and mobile computing; • **Computing methodologies** → Artificial intelligence.

Keywords

Large Language Models, On-device Inference, Multitasking, Memory Pressure



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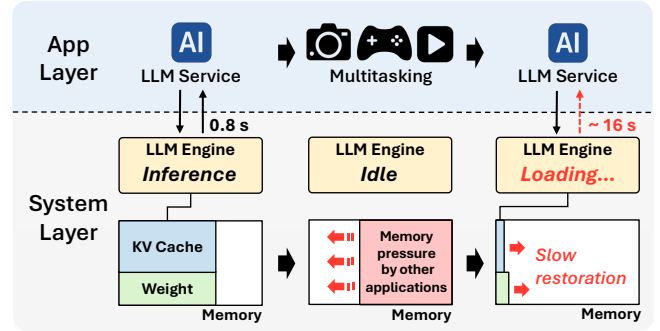


Figure 1: Multitasking on mobile devices: the user interacts with an on-device LLM service, switches to other apps, and later returns. Memory pressure from concurrent apps evicts LLM memory, increasing the response time from 0.8s to 16s.

1 INTRODUCTION

Large Language Model (LLM)-based services are rapidly gaining adoption on mobile devices [2, 6, 24, 43], unlocking personalized experiences that leverage local user data without cloud dependence. On-device LLM assistants, for example, can summarize email threads using local message history or suggest contextually relevant responses based on prior conversations [10, 33, 71]. These benefits stem from preserving the *context* of prior interactions.

LLM inference systems achieve this by storing the interaction history as a set of key-value pairs known as the *KV cache*. Preserving the KV cache enables generating context-aware responses to new inputs while avoiding expensive recomputation. Despite its benefits, as the KV cache accumulates over long or repeated interactions, even a single context for a small mobile-optimized model can grow to around 4 GB. When combined with *model weights*, the total LLM memory footprint reaches 5 to 7 GB, consuming nearly half of the typical 12 GB RAM available on modern smartphones [5, 41, 44].

Such large memory demand poses a critical challenge in mobile *multitasking environments*, where users frequently switch between multiple applications. As illustrated in Figure 1, after an LLM interaction completes, users transition to other applications such as camera, game, or video streaming.

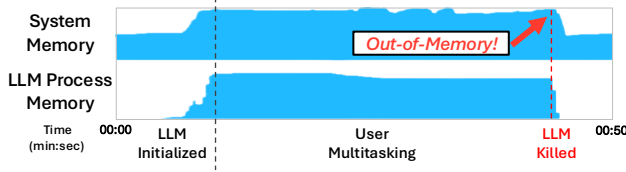


Figure 2: System and LLM process memory traces under multitasking, collected on a Galaxy S25+ using Perfetto [4]. As memory pressure increases, the Low-Memory Killer terminates the LLM process.

As more apps are launched and used, the limited memory of mobile devices becomes rapidly scarce. Under this memory pressure, the operating system (OS) evicts the idle LLM memory to make room for active applications. When the user returns to the LLM service with a new inference request, this evicted memory must be restored before a response can be generated. During this restoration, inference computation is blocked by slow storage reads and on-demand paging mechanisms [48], substantially increasing Time-to-First-Token (TTFT)—the time to generate the first output token—from 0.8 to 16 seconds.

To overcome the limited bandwidth of storage-only restoration, mobile OSes such as Android employ zRAM [3], a compression-based in-memory swap space that compresses evicted pages within RAM itself. However, zRAM’s general-purpose compression algorithms (e.g., lz4 [13] and zstd [14]) barely compress the KV cache, leaving memory pressure largely unrelieved. As shown in Figure 2, this unresolved pressure leads to process termination by the Low-Memory Killer. As a result, before generating a new response, the LLM system must reload weights from storage and recompute the entire KV cache. Both operations incur significant latency, severely degrading the responsiveness and user experience.

As such, general OS memory management mechanisms are largely unsuitable for LLMs. Unfortunately, existing mobile LLM services [53, 60, 65] focus primarily on runtime inference optimizations, and their memory management is entirely dependent on these OS-based mechanisms. As a result, the TTFT and user experience of LLM-based applications suffer greatly in multitasking mobile environments, calling for an LLM- and mobile-aware system that properly handles external memory pressure.

To this end, we present *mzCache*, a complete LLM inference system that integrates specialized memory management for responsive inference in mobile multitasking environments. A key challenge is that memory pressure from competing applications is inherently dynamic and unpredictable. To handle this, *mzCache* employs a restoration-oriented design, elastically reducing its memory footprint according to external pressure while maintaining readiness for rapid restoration throughout the eviction process.

mzCache first establishes a memory management mechanism tailored to mobile platforms: it partitions weights and KV cache into fine-grained units and allocates them as shared buffers on the mobile SoC’s unified memory. This enables evicting only the necessary amount of memory while allowing the CPU to perform restoration and the GPU to execute inference without redundant data transfers between processors. On top of this, *mzCache* introduces two policies that govern where and when each unit should be evicted, so that any partial eviction state allows for the fastest possible restoration regardless of the eviction amount.

To overcome the storage read bottleneck in restoration, we draw inspiration from Android’s zRAM and applies KV cache compression techniques [23, 36, 37, 47, 68] that exploit its structural properties, making in-memory compressed space a viable second restore path alongside storage reads. Evicted KV cache is distributed across the two paths so that both remain fully utilized during restoration. Beyond placement, *mzCache* evicts later layers first so that early layers, which LLM inference executes first, are always preserved. During restoration, the GPU can begin inference immediately on the preserved front layers while the CPU restores evicted layers in the same forward order, maximizing the overlap between restoration and inference.

We implement *mzCache* on top of llama.cpp [18], a widely used LLM inference framework for mobile environments, and evaluate its performance across various mobile devices and LLM models. Compared to a storage-backed partial offload baseline, *mzCache* reduces TTFT by 2.1–5.5× and demonstrates effectiveness in real-world multitasking scenarios under memory pressure. The main contributions of this paper are summarized as follows:

- We reveal that OS-managed memory mechanisms in mobile multitasking environments are inadequate for LLM workloads, causing severe TTFT degradation.
- We present *mzCache*, a complete LLM inference system with restoration-oriented memory management that enables responsive inference from any eviction state.
- We test *mzCache* on commercial smartphones, show 2.1–5.5× faster TTFT than partial offload, and validate its practicality under realistic multitasking pressure.

2 BACKGROUND

2.1 Characteristics of LLM

Modern LLMs are constructed by stacking dozens of transformer layers [57]. During inference, input tokens are processed through these stacked layers to generate output tokens. The inference consists of two stages: First, the *prefill* stage processes the entire input sequence of tokens. For each token, every transformer layer computes key and value (KV)

Table 1: Memory footprint of representative LLMs used in mobile and edge environments. All models listed here have fewer than 3 billion parameters.

LLM model	Weight	Context Length (max)	KV (max)	Max footprint (Weight + KV)
EXAONE4-1.2B [32]	2.6 GB	64k	3.8 GB	6.4 GB
Gemma2-2B [52]	4.9 GB	8k	0.8 GB	5.7 GB
Llama3.2-1B [38]	2.5 GB	128k	4 GB	6.5 GB
Qwen3-0.6B [54]	1.2 GB	32k	3.5 GB	4.7 GB
Qwen3-1.7B [54]	3.5 GB	32k	3.5 GB	7.0 GB

vectors, and the resulting KV pairs are stored as *KV cache*. Because self-attention in this stage requires interactions across all input tokens, the prefill latency—referred to as Time-to-First-Token (TTFT)—grows quadratically with the input sequence length. The second stage, called the *decode* stage, autoregressively generates output tokens one by one. In the decode stage, each newly generated token references the accumulated KV cache to avoid recomputation, making each decoding step far cheaper than prefill.

2.2 Context-Aware LLM Inference in Mobile

In mobile LLM inference, preserving the KV cache across interactions is particularly important, as applications are designed to deliver personalized experiences. For instance, mobile LLM services process private user data—such as message history, email threads, and chat logs—as inference context, storing it as KV cache [10, 33, 71]. The model then references this KV cache during inference to generate relevant responses across interactions.

Preserving the KV cache, however, directly affects the memory footprint. As shown in Table 1, for models under 3 billion parameters, the KV cache alone can grow up to 4 GB depending on context length, reaching 5–7 GB when combined with model weights. In practice, reaching such long contexts is common: popular LLM datasets show that a single interaction produces hundreds of tokens on average [45, 63, 69], and contextual input such as email threads easily exceeds a thousand tokens [61]. As these interactions accumulate throughout the day, the context quickly approaches its maximum.

The large KV cache also affects the computational cost of prefill. When a new query arrives, the model must process all new input tokens while performing attention over the entire stored KV cache. This causes the prefill computation to scale with both the new input length and the size of the accumulated KV cache, thus increasing the TTFT.

2.3 Memory Pressure under Multitasking

Modern mobile devices operate in highly dynamic multitasking environments, where users switch between applications

Table 2: KV cache space savings under lz4 and zstd at an 8k-token WikiText context. These zRAM algorithms achieve only 0.2–9.3% space savings, indicating KV cache compresses poorly under zRAM.

	EXAONE4 1.2B [32]	Gemma2 2B [52]	Llama3.2 1B [38]	Qwen3 0.6B [54]	Qwen3 1.7B [54]
lz4 (%)	0.28	0.2	1.19	0.19	0.19
zstd (%)	9.30	8.65	9.28	8.04	8.04

over 100 times per day [16]. As multiple apps run concurrently, they create unpredictable memory pressure on the limited RAM available on mobile devices. Under such pressure, the mobile operating system (OS) must reclaim memory from inactive processes, such as background applications that the user is not actively using, to make room for active applications [3].

To reclaim memory under pressure, the OS relies on two mechanisms. The first is *paging*, which evicts pages from RAM: file-backed pages, whose original contents reside on storage, are simply dropped, while anonymous pages, which have no backing on storage, are swapped out to a swap space. If the swap space resides in flash storage, as a swap file, the high latency of storage reads during restoration severely degrades responsiveness. To address this, modern mobile OSes such as Android employ zRAM [3]—an in-memory compressed swap space that stores evicted pages in compressed form within RAM itself, trading CPU cycles for faster restoration compared to storage-backed swap. When memory pressure persists despite paging, the OS escalates to its second mechanism: the *Low-Memory Killer (LMK)*, which terminates background processes to free their entire memory.

3 MOTIVATION

3.1 Limitations of OS Memory Management for LLMs

Depending on how the inference engine allocates memory, LLM data resides in one of two memory regions—*pageable* or *page-locked*—and under multitasking memory pressure, the OS handles neither region well.

1) *Pageable region*. The OS can reclaim this region through paging, but with mechanisms blind to LLM data. For LLM processes, file-backed pages containing model weights are dropped, while anonymous pages containing the KV cache are written to zRAM. However, zRAM uses general-purpose compression algorithms such as lz4 [13] or zstd [14], which rely on byte-level pattern matching. These algorithms fail on KV cache because it consists of arithmetic data that lacks such patterns: as shown in Table 2, space savings range from only 0.2% to 9.3% across different LLM models. As a result, memory pressure remains largely unrelieved, as the

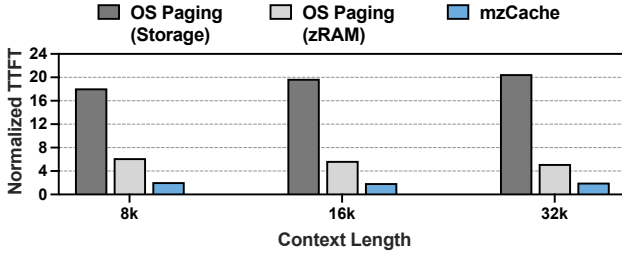


Figure 3: TTFT under full eviction (Galaxy S25+, Qwen3-0.6B, 8 input tokens), normalized to the all-in-memory case. mZCache keeps TTFT close to the in-memory case. Note that zRAM still leaves memory pressure largely unrelieved (Table 2).

poorly compressed KV cache continues to occupy substantial memory in zRAM. When such pressure persists, the LMK eventually terminates the LLM process, whose large memory consumption makes it particularly vulnerable [1]. Upon termination, the entire KV cache representing the user’s context is lost. Resuming the service then requires a cold-start that reloads weights from storage and recomputes the entire KV cache from scratch, leading to severe responsiveness degradation.

Even when the LLM process avoids termination, the restoration introduces substantial latency. When the user returns, the OS must restore both weights from storage and KV cache from swap space through on-demand paging: pages are brought back into memory only when accessed during inference, triggering page faults [48]. Because prefill must access all weights and the entire KV cache to generate the first token, page faults continue throughout restoration until all evicted data has been recovered. As shown in Figure 3, this on-demand restoration severely degrades TTFT. Under full eviction, TTFT increases by 18–20 times when using flash storage swap and 5–6 times when using zRAM, compared to the all-in-memory case.

2) *Page-locked region*. Memory pinned by an accelerator (GPU or NPU) driver, or locked by an application through system calls such as `mlock`, is exempt from paging: the OS cannot evict these pages regardless of pressure. The only lever the OS retains for this region is the LMK—terminating the process—which incurs the cold-start penalty described above.

To avoid even this penalty, Android AICore [2] blocks both reclamation mechanisms: it locks LLM weights in memory to exempt them from paging [55, 56], and provides Google’s Gemini Nano model as a system-level service, which is effectively shielded from LMK termination [3]. As a result, the OS is left with practically no means to reclaim this memory. This protection, however, comes at a critical cost for mobile

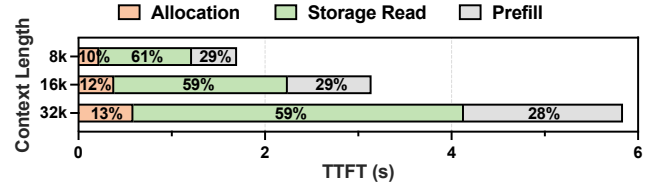


Figure 4: TTFT breakdown when loading entire weights and KV cache from flash storage and executing inference on GPU (OnePlus 12, Qwen3-0.6B, 8 input tokens), showing time spent on allocation, storage reads, and prefill across different context lengths.

devices: the locked weights occupy RAM even when the service is idle, wasting scarce memory and reducing headroom for other applications. Moreover, it provides no solution for the KV cache, which can still be evicted under pressure, reintroducing significant latency upon resumption.

3.2 Design Constraints

Since the OS handles neither memory region well, *mobile LLM inference systems must themselves incorporate principled memory management*. Building such a solution requires two capabilities: (1) evicting memory when other processes create pressure, making room for concurrent applications, and (2) rapidly restoring evicted data and serve inference when a new request arrives. However, several constraints in existing LLM systems and mobile platforms make achieving these capabilities challenging.

Monolithic memory buffer. LLM inference systems typically adopt monolithic memory allocation: they allocate one large contiguous buffer at initialization to avoid repeated allocation overhead and simplify memory layout management for performance-critical kernels. However, the monolithic approach operates as all-or-nothing: the system cannot adjust its memory footprint in response to memory pressure. Without mechanisms to partially release or allocate memory, adaptive memory management becomes infeasible.

Limited memory hierarchy. On mobile platforms, prefill relies on accelerators such as GPUs to achieve acceptable performance [11, 61]. When data is not resident in accelerator memory, the system must restore it before prefill can proceed. On servers, discrete GPU systems provide a separate host memory tier: data offloaded from GPU memory can be temporarily stored in host memory and quickly restored via high-bandwidth interconnects such as PCIe [12, 17, 25, 67]. Mobile platforms, however, employ unified memory architectures where accelerators and the CPU share the same physical memory, eliminating this intermediate fast-access tier. Consequently, systems must rely on flash storage, where slow read performance becomes a critical bottleneck. As shown in

Figure 4, storage reads dominate restoration time, accounting for 59–61% of TTFT across different context lengths.

Allocation overhead. Since eviction releases memory regions back to the system, restoration requires re-allocating them before any data can be loaded. This re-allocation incurs substantial overhead, as it involves system calls that trigger page allocation, zeroing, and IOMMU / SMMU¹ page table updates [7, 22]. The cost scales with the allocation size and recurs at every restoration. As shown in Figure 4, allocation accounts for 10–13% of TTFT, representing a non-trivial cost that compounds the restoration latency.

3.3 Opportunities

To achieve the goal of responsive LLM inference restoration, it is critical to understand and leverage the unique characteristics of mobile memory systems. We identify two such characteristics: (1) the use of compressed memory regions, and (2) the unified memory architecture.

Due to the low access speeds of flash storage on mobile devices, memory offloading on mobile is often bound by storage I/O. To overcome this, Android’s zRAM offers a compressed region in the main memory, providing a small but high-throughput alternative to the larger but slower storage path (§3.1). We reinterpret this dual-choice offloading concept of the mobile OS for LLM memory management, and apply compressed offloading to the dynamically generated KV cache while keeping the static model weights on the flash storage. The reason is clear: the KV cache exhibits similarity between adjacent tokens and channels, allowing higher compression ratios with minimal accuracy loss [23, 36, 37, 47, 68], while the larger model weights can remain unchanged in storage, allowing reduced I/O on eviction and a more efficient use of the system’s memory resources.

The single shared physical memory on mobile devices also provides a key opportunity for efficient LLM memory management. This unified memory provides a common workspace for different processors of the mobile SoC to coordinate the use and management of the LLM memory, often at the same time. For example, the GPU can focus on high-throughput prefill computation, while the CPU handles memory management tasks such as allocation, storage reads, and decompression, without requiring costly memory copies between the two processors. By carefully partitioning the LLM memory into manageable chunks and scheduling the access patterns between the GPU and CPU, the two processors can cooperate to leverage the unified memory of the mobile system to its full potential.

¹IOMMU (I/O Memory Management Unit) and SMMU (System MMU) translate device virtual addresses to physical addresses for accelerators such as GPUs and NPUs.

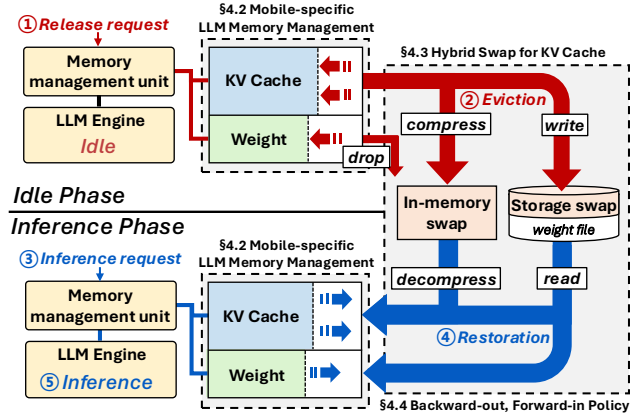


Figure 5: The System Overview of mzCache.

A natural concern in exploiting both opportunities simultaneously is memory bandwidth contention, as the two restore paths and the two processors all share the physical memory subsystem. In practice, however, this shared bandwidth is an underutilized resource, since no single processor can fully utilize it on its own [11] and storage reads occupy only a small fraction of it². The shared memory bandwidth is thus an enabler rather than a bottleneck: engaging the CPU, GPU, and storage together puts this spare bandwidth to work for restoration.

4 SYSTEM DESIGN

4.1 Overview

Figure 5 presents an overview of *mzCache*. *mzCache* operates in two distinct phases: the *idle phase* and the *inference phase*. During the idle phase, when the system receives a memory release request due to external memory pressure ①, *mzCache* evicts the requested amount of LLM memory ②. The idle phase continues until a new inference request arrives ③. During the inference phase, *mzCache* restores evicted LLM memory ④ and performs inference ⑤. After the inference completes, the system transitions back to the idle phase.

Unlike server-side LLM systems, which have exclusive control over GPU memory, a mobile LLM inference system faces *dynamic and unpredictable* pressure from competing user applications. To overcome this novel challenge, it is critical that LLM memory is elastically reduced according to the outside memory pressure, evicting only the necessary amount of memory while maintaining readiness for rapid restoration throughout the eviction process. *mzCache* enables and leverages this **Restoration-oriented Eviction** concept to the fullest, allowing low-latency restoration of

²On Galaxy S25+, the sequential-read bandwidth of UFS 4.0 (4.2 GB/s) is far below the peak bandwidth of LPDDR5X (84.8 GB/s).

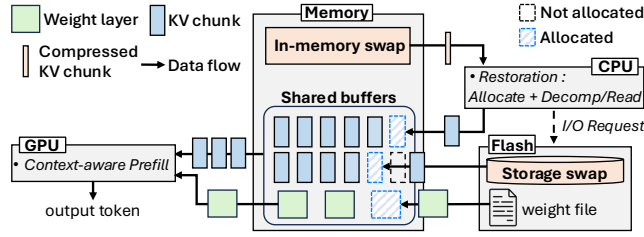


Figure 6: Overview of GPU-CPU cooperation on the mobile SoC's unified memory during mzCache restoration.

LLM memory under any pattern or degree of memory pressure that an LLM inference system may face in the mobile environment.

To realize this, we first design an LLM memory management mechanism that can leverage the unified memory of mobile environments to enable fine-grained management of LLM memory with concurrent model execution (§4.2). On top of this mechanism, two policies govern the eviction and restoration process to ensure efficient memory recovery from any intermediate eviction state, providing a robust solution to the problem of dynamic memory pressure in the mobile environment. The first policy determines *where* to place evicted data across the two restore paths, in-memory decompression and storage reads, to balance restoration load between the two paths (§4.3). The second policy determines *when* each LLM data should be evicted and restored, to minimize the time to first token generation in LLM inference restoration (§4.4).

4.2 Mobile-specific LLM Memory Management

The main goal of mzCache's memory management mechanism is to bridge the gap between mobile memory systems and its restoration-oriented eviction policies. To achieve this, we first depart from the monolithic allocation of traditional LLM inference systems and partition the layer weights and KV cache into finer-grained units, allowing the system to elastically evict data only to the minimum required amount. We use layer-level granularity for weights, since each layer forms a fixed-size unit that serves as a natural boundary for model execution. For KV cache, the layer-level granularity is insufficient as the KV cache size varies with context length. We instead divide each layer's KV cache into fixed-size chunks following popular server-side practices [28], enabling flexible partial eviction regardless of context length. We detail the exact chunk size of the KV cache in §4.3, as the most effective size depends on the I/O characteristics of the given memory system.

Using Figure 6 as an example, we show how mzCache leverages the unified memory and fine-grained units to achieve low-latency LLM memory restoration on mobile devices. We first allocate the weight layers and KV cache chunks (KV chunks) as *shared buffers* directly visible to both the CPU and the GPU. When a new inference request arrives after a partial eviction, mzCache immediately begins the prefill computation of the request on the GPU using the partial weights and KV chunks that remain in shared buffers. Concurrently, mzCache also restores partially evicted data on the CPU by loading weight layers and KV chunks from storage via direct I/O and decompressing KV chunks from in-memory swap, all directly into shared buffers. Since each fine-grained memory unit can be independently allocated and loaded, we distribute memory allocation and data loading across multiple CPU cores to maximize restoration throughput. This design removes any explicit data transfers between the two processors and allows seamless pipelining between the restoration on the CPU and the consumption of the restored data on the GPU to achieve minimum-latency LLM memory restoration in mobile environments.

To compute directly on KV chunks residing in non-contiguous shared buffers, mzCache employs a custom attention kernel using OpenCL [27]. Unlike server-side systems that must coordinate KV cache history across concurrent multi-user requests using a prefix-tree system [28, 67, 70], mobile inference serving a single user can manage KV cache through a simple array of KV chunk pointers. Our kernel targets prefill over a large KV cache accumulated from prior interactions. Due to the restricted load/store bandwidth of mobile GPU on-chip memory [29] and limited data reuse from single-batch serving of mobile LLMs, we find that caching the KV data on the GPU on-chip memory is often slower than streaming directly from shared buffers. Therefore, the kernel reads KV chunks directly from shared buffers and streams through the chunk pointer array, applying online softmax [15] across non-contiguous chunks to enable seamless single-pass attention without consolidation. To maximize throughput, the kernel employs 8-wide FP16 vector operations that exploit the double-rate FP16 execution hardware of the Qualcomm Adreno GPUs that we mainly target.

4.3 Hybrid Swap for KV Cache

As identified in §3.3, compressing the KV cache in memory during eviction opens a second restore path—in-memory decompression—alongside storage reads. However, a fixed assignment that simply places the KV cache in in-memory space and the weights on storage is insufficient: the two restore paths have different throughputs and the KV cache size varies with context length, so restoration inevitably becomes imbalanced. To address this, mzCache introduces

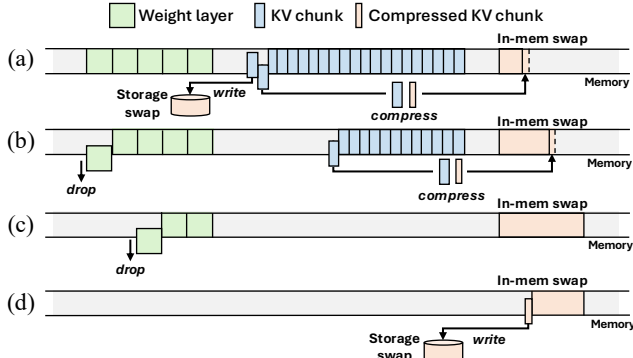


Figure 7: Four eviction stages in mzCache. (a) KVONLY: evict KV chunks to both swap spaces. (b) KVANDW: drop weights while evicting KV. (c) WONLY: drop only weights. (d) COMPKV: compressed KV may be evicted to storage under further pressure.

hybrid swap for the KV cache: rather than assigning KV cache exclusively to in-memory compressed space, a portion is also written to storage, enabling the system to distribute data across both paths.

The key question is how to determine this distribution, given that the system cannot know in advance how much memory will ultimately be evicted. Consider, for example, a policy that evicts all KV chunks first across both paths before dropping any weight layer. Distributing KV cache evenly may appear balanced initially. But once all KV cache has been evicted and further eviction is needed, the system has no choice but to drop weight layers, which must later be reloaded from storage. As more weight layers are dropped, the restoration load shifts increasingly toward the storage path, breaking the balance. This illustrates that a more principled approach is needed.

Offline profiling. Balancing the two restore paths requires knowing how long each takes to process a unit of data. These per-unit times come from a one-time profiling on each target device, which measures two throughputs: the decompression throughput of the configured compression algorithm and the sequential read throughput of the flash storage via direct I/O. Given the sizes of a KV chunk and a weight layer, dividing each by the corresponding throughput yields the three per-unit times: t_d^{KV} (decompress one KV chunk), t_r^{KV} (read one KV chunk from storage), and t_r^W (read one weight layer from storage). Even when a different model changes these unit sizes, the per-unit times are simply re-derived from the same measured throughputs, without re-profiling the device.

KV chunk size. For the per-unit profiling above to be valid, each path must operate at full throughput even for a single unit. Decompression satisfies this regardless of data size—in

Algorithm 1 Hybrid Swap Eviction Policy

Input: Per-unit times t_d^{KV} , t_r^{KV} , t_r^W ; threshold T

```

1:  $N_{KV} \leftarrow N_{KV}^{total}$   $\triangleright$  KV chunks not yet evicted
2:  $N_W \leftarrow N_W^{total}$   $\triangleright$  weight layers not yet dropped
3: while release request do
4:   if  $N_{KV} > T$  then
5:      $N_{KV} \leftarrow KVENLY(N_{KV}, t_d^{KV}, t_r^{KV})$ 
6:   else if  $N_{KV} > 0$  and  $N_W > 0$  then
7:      $N_{KV}, N_W \leftarrow KVANDW(N_{KV}, N_W, t_d^{KV}, t_r^W)$ 
8:   else if  $N_W > 0$  then
9:      $N_W \leftarrow WONLY(N_W)$ 
10:  else
11:     $COMPKV()$ 
12:  end if
13: end while

```

our measurements, its throughput remains constant. Storage reads, in contrast, reach full bandwidth only when each read request is sufficiently large: in our experiments on UFS 4.0 flash storage, sequential reads saturate the storage bandwidth at approximately 512 KB per operation. Weight layers, at tens of megabytes each, are well above this size. To meet this threshold for models with a small KV hidden dimension (e.g., 512), a minimum of 256 tokens per chunk is required.³ We therefore fix the chunk size at 256 tokens across all configurations.

Four-stage eviction. Since KV chunks can be directed to either restore path, they serve as the balancing lever between the two paths. To keep the two paths balanced, mzCache sets aside enough KV chunks to cover the weight reload: the threshold T is the number of KV chunks that keeps the decompression path occupied while all weight layers are reloaded from storage. The eviction policy follows from how the remaining number of KV chunks (N_{KV}) relates to T throughout eviction.

Let N_{KV}^{total} and N_W^{total} denote the total number of KV chunks and evictable weight layers, respectively.⁴ While reloading one weight layer from storage, the decompression path can process $\lfloor t_r^W / t_d^{KV} \rfloor$ KV chunks, giving:

$$T = N_W^{total} \cdot \lfloor t_r^W / t_d^{KV} \rfloor$$

Two cases arise based on whether N_{KV}^{total} exceeds T (Algorithm 1, Figure 7).

1) **KVENLY** $\rightarrow$ **KVANDW** ($N_{KV}^{total} > T$). The KV cache exceeds what the decompression path alone can handle, so eviction starts in KVENLY, evicting KV chunks to both swap spaces while weights remain untouched. The chunks are split in

³ $256 \times 512 \times 2$ (K,V) $\times$ 2B (FP16) = 512KB.

⁴ We set N_W^{total} to the total number of weight layers, allowing full eviction; setting it to 0 would keep all weights resident in memory.

inverse proportion to t_d^{KV} and t_r^{KV} —the faster path taking more—so that both paths finish restoring their shares in the same amount of time. Once N_{KV} falls to T , the system transitions to KVANDW, compressing remaining KV chunks to in-memory space while dropping weight layers—for every layer dropped, $\lfloor t_r^W / t_d^{KV} \rfloor$ KV chunks are compressed. Because N_{KV} equals T at this transition, the remaining KV chunks and the weight layers run out together.

2) $KVANDW \rightarrow WONLY$ ($N_{KV}^{total} \leq T$). All KV cache fits within the decompression path, so eviction starts directly at KVANDW. With fewer KV chunks than the weight reload requires, however, the KV cache is exhausted while weight layers still remain. Eviction then continues in *WONLY*, dropping the remaining weight layers.

In either case, once all weight layers have been dropped and all KV chunks evicted, only the compressed KV chunks remain in the in-memory swap space. If memory pressure persists even then, the system enters *COMPKV*, moving the compressed KV to storage to release this last portion of memory.

Balance preservation. Throughout *KVONLY* and *KVANDW*, every increment of eviction adds equal restoration work to both paths, so the placement remains balanced wherever eviction stops. *WONLY* and *COMPKV* extend beyond the point where this balance can be maintained: no KV chunks are left to assign to the decompression path, so any further eviction can no longer be balanced across the two paths. Because they are entered only after the earlier stages have balanced the placement across both paths, the system still preserves the best achievable balance for the given eviction amount.

4.4 Backward-out, Forward-in Policy

While §4.3 determines where to place evicted data across restore paths, this section addresses in what order to evict and restore weight layers and KV chunks. To determine this order, we must consider the sequential dependency structure of LLM inference: each layer (L_i) cannot begin until the preceding layer (L_{i-1}) completes, so eviction ordering directly determines whether restoration and prefill can proceed concurrently. Consider, for example, an LRU-based policy, widely used in OS page reclamation [20], that evicts the earliest-accessed layers starting from L_1 . Even if only a small amount of memory is reclaimed, prefill cannot start until L_1 is restored, leaving the GPU entirely idle. Under this policy, prefill is always blocked by restoration regardless of the eviction amount. This is especially problematic in mobile environments where the eviction amount is unpredictable.

mzCache takes the opposite approach, following the well-known MRU-style ordering (Figure 8a). The system evicts in reverse layer order, always taking the rearmost of the

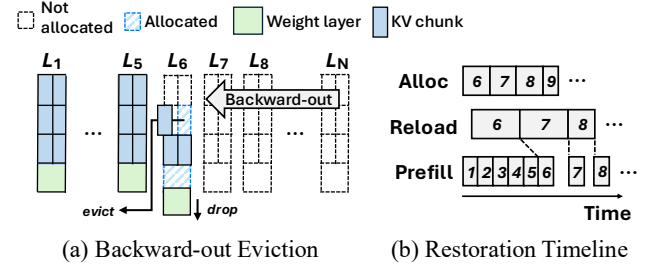


Figure 8: Backward-out eviction and restoration overlap in mzCache. (a) Eviction proceeds from the last layer backward. (b) Timeline of GPU prefill overlapping with CPU restoration of evicted layers.

remaining weight layers and KV chunks.⁵ This *backward-out* eviction ensures that early layers are always preserved regardless of how much memory is ultimately reclaimed. Restoration then proceeds forward from the earliest evicted layer (*forward-in*), matching the order inference will access them. As prefill progresses through $L_1 \rightarrow L_2 \rightarrow \dots$, the CPU concurrently restores evicted layers in the same direction, naturally overlapping the two processes and minimizing prefill stalls (Figure 8b).

The backward-out, forward-in policy produces a natural LIFO access pattern: compressed KV chunks are added in tail order during eviction and removed in head order during restoration. mzCache exploits this by organizing the in-memory swap space as a stack. Rather than pre-allocating a large fixed-size stack, the system begins with a small stack and grows incrementally—allocating additional stack segments only when needed and releasing them after restoration completes. This approach minimizes the memory footprint and reduces fragmentation.

5 IMPLEMENTATION

We implement mzCache with 6k lines of C/C++ on top of llama.cpp [18], an open-source LLM inference framework that supports various backends including mobile GPUs [58]. **Compression and kernels.** mzCache treats KV cache compression as a modular component with a drop-in replaceable interface. In our implementation, we adopt two existing compression algorithms: 8-bit quantization [47] and CacheGen [36], a higher-ratio technique at the cost of slower decompression. We use 8-bit quantization as the default method, while a comparison between the two is presented in §6.5. To enable these compression methods on the CPU, we implement new compression and decompression kernels using ARM NEON SIMD instructions [8], targeting the

⁵Each layer holds (context length)/256 KV chunks, each organized hidden-dimension-first.

Table 3: Evaluation testbed specifications

Devices				
Device	CPU	GPU	Memory	Storage
Galaxy S25+	2×Oryon Phoenix L 6×Oryon Phoenix M	Adreno 830	12GB LPDDR5X	UFS 4.0
OnePlus 12	1×Cortex-X4 5×Cortex-A720 2×Cortex-A520	Adreno 750	12GB LPDDR5X	UFS 4.0
LLM Models				
Model	Weights	KV cache (8k / 16k / 32k)		
Qwen3-0.6B	1.2 GB	0.9 GB / 1.8 GB / 3.5 GB		
EXAONE-4.0-1.2B	2.6 GB	0.5 GB / 1.0 GB / 1.9 GB		

ARMv8-A architecture commonly found in mobile devices. Our custom GPU attention kernel is implemented in 0.6k lines of OpenCL [27], and shared buffers between the CPU and GPU are realized using OpenCL’s Shared Virtual Memory (SVM) [9].

Eviction and restoration interface. mzCache provides interfaces for both eviction and restoration operations, assuming that the amount of memory to release is provided externally:

- `evict(size)`: Given the amount of memory to be released, this function evicts LLM memory based on the current state until the specified size is freed.
- `restore_generate(input_tokens)`: This function restores evicted LLM memory based on the current state. Using C++ primitives such as `std::promise`, it allows the GPU to begin prefill computation on layers as soon as they are restored.

System integration. SVM buffers are allocated through the GPU driver and thus belong to the page-locked region described in §3.1, which the OS paging path cannot reclaim. Consequently, mzCache operates independently of OS paging path, and deploying mzCache requires no modifications to the OS kernel. For memory pressure detection, we use Android’s `onTrimMemory` callback in our application-level deployment (§6.6).

6 EVALUATION

This section evaluates mzCache, analyzing the impact of its optimization techniques and demonstrating its practicality in mobile multitasking environments. We first compare overall performance against baselines (§6.2), then dissect the contribution of each core technique (§6.3), measure power and energy consumption (§6.4), compare compression algorithms in terms of compression ratio, accuracy, and TTFT (§6.5), and validate its effectiveness in real-world deployment (§6.6).

6.1 Experimental Setup

Devices. Table 3 summarizes device specifications. We evaluate on two off-the-shelf smartphones representing different processor generations. On Galaxy S25+ [43], big cores handle decompression and storage I/O while middle cores perform allocation. On OnePlus 12 [40], we distribute workloads across the six higher-performance cores, excluding the two little cores.

LLM models. We evaluate mzCache using two recently released lightweight LLMs: Qwen3-0.6B [54] and EXAONE-4.0-1.2B [32]. Table 3 shows their weight sizes and KV cache sizes at different context lengths.

Number of tokens. Real-world LLM conversation traces show that accumulated context readily reaches thousands of tokens as interactions progress [63]. We use context lengths of 8k, 16k, and 32k tokens to evaluate performance across representative KV cache sizes. We fix the input sequence length to 8 tokens to model a short follow-up query against the accumulated context.

Baselines. We compare mzCache against two baselines:

1) *OS Paging.* We leverage the default Android paging mechanism to represent the behavior of LLM services that rely solely on OS memory management. Model weights are managed via standard memory mapping (`mmap`) as file-backed pages, while the KV cache is treated as anonymous memory and compressed into zRAM using the `zstd` algorithm [14]. Since this approach relies on OS-level page fault handling, we run inference on the CPU: GPU buffers reside in the page-locked region (§3.1), which the OS paging path cannot reclaim, so a GPU-based deployment offers no graceful reclaim path and thus cannot serve as a partial-eviction baseline. On the CPU, pages are restored on demand as they are accessed.

2) *Partial Offload.* Since no existing mobile LLM system provides mechanisms to handle memory pressure, we implement a partial offload baseline in `llama.cpp`. Note that default `llama.cpp` adopts monolithic memory allocation, limiting it to all-or-nothing eviction. To enable fair comparison with mzCache across different eviction amounts, we extend it to support partial eviction and restoration at layer granularity for both weights and KV cache. Upon restoration, the baseline sequentially loads weights and KV cache from storage into GPU memory before executing prefill on the GPU. Consequently, the eviction and restoration order does not affect its performance. Only the eviction amount matters.

Measurements. We measure CPU memory usage via `smaps` and GPU memory usage through the `gpumem_mapped_bytes` value exposed by the Qualcomm KGSL driver [42]. `smaps` reports how much data is swapped out, not how much space it occupies after zRAM compression. Thus, for OS Paging,

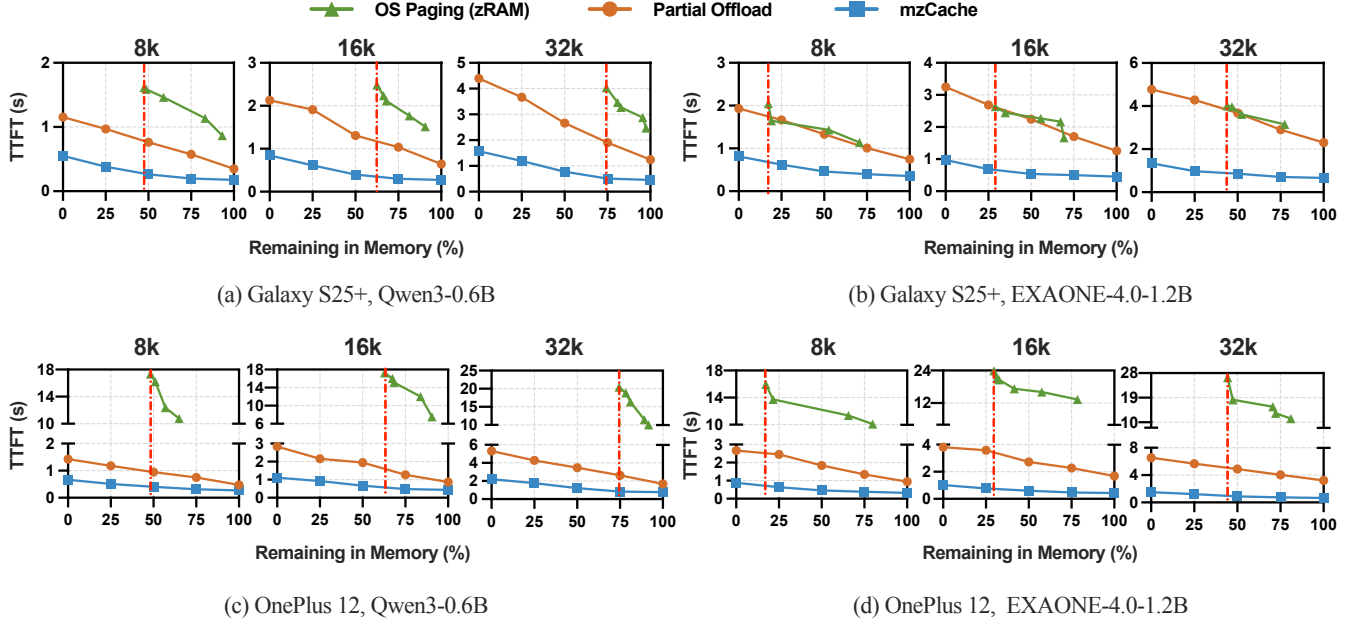


Figure 9: TTFT across different remaining-memory levels. The red dashed line marks full eviction under OS Paging, which still retains non-negligible memory due to poor KV cache compression. mzCache consistently achieves lower TTFT than baselines across all conditions.

we pre-profile the compression ratio from full swap-out scenarios and apply it to the reported size to estimate the actual memory footprint. Power consumption is estimated by multiplying the battery’s current and voltage values obtained from sysfs. To evaluate the impact of different compression algorithms on accuracy, we measure F1 scores using 428 question-answer pairs from 30 documents in TriviaQA [26], whose context lengths range from 8.8k to 28k tokens. Except for application-level deployment experiments, all measurements are conducted through Android Debug Bridge (ADB) [19].

6.2 Overall Performance - TTFT

We evaluate restoration and inference scenarios with varying amounts of LLM memory remaining in system memory. Remaining in memory denotes the percentage of weights and KV cache that stays resident in memory after eviction. This includes compressed KV cache held in system memory. We exclude token embeddings for simplicity.

For Partial Offload and mzCache, we compare the restoration performance at 0%, 25%, 50%, and 75% of remaining memory. Since granularity differs between the two, we allow a $\pm 3\%$ margin, always evicting slightly more in mzCache to ensure fairness. For OS Paging, we evict data by running a memory-intensive dummy process to induce OS-level reclamation. Since the ratio of evicted weights to KV cache at a given remaining memory level can affect TTFT, we drop

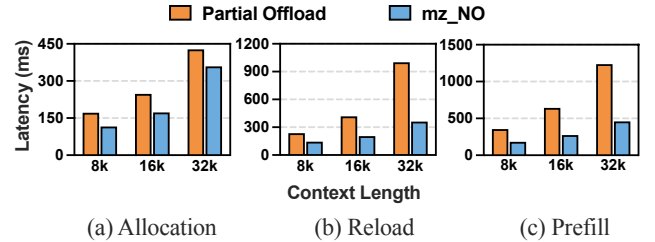


Figure 10: Latency breakdown for (a) allocation, (b) reload, and (c) prefill. Partial Offload vs. mzCache without overlap (mz_NO).

weights before swapping out KV cache to keep this ratio fixed across runs, isolating the remaining-memory level as the sole variable.

TTFT comparison. Figure 9 shows that mzCache consistently achieves lower TTFT than Partial Offload across all configurations. Specifically, mzCache achieves $2.1\times$ – $5.5\times$ speedup at 0%, 25%, 50%, and 75% remaining memory levels. The speedup is consistent regardless of KV cache size, model type, or device, demonstrating mzCache’s effectiveness across diverse scenarios. OS Paging performs worse than Partial Offload in most cases. Consistent with Figure 3, this latency is dominated by on-demand paging during restoration rather than prefill computation. On the OnePlus 12, OS

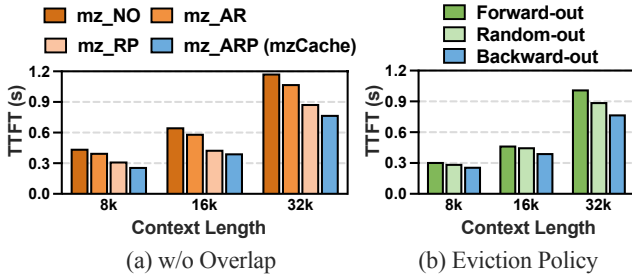


Figure 11: TTFT with different (a) overlapping strategies and (b) eviction policies. **mz_NO**: no overlap, **mz_AR**: Allocation-Reload overlap, **mz_RP**: Reload-Prefill overlap, **mzCache**: Allocation-Reload-Prefill overlap.

Paging exhibits significantly higher latency with large variance. During restoration, we monitored the resident set size (RSS) and observed that it sometimes decreased rather than increased, indicating that on-demand page faults triggered additional evictions of already-resident pages. This cascading effect substantially increases TTFT.

Analysis of compressed KV cache. The red vertical dashed line in Figure 9 marks the remaining memory when all weights are dropped and all KV cache is swapped out under OS Paging. Even with full eviction, OS Paging consumes substantial memory due to poor KV cache compression. In contrast, mzCache reduces remaining memory to 0% by evicting compressed KV cache to storage in the CompKV stage. At these respective full-eviction states, mzCache achieves a $2.5\times$ – $3.0\times$ speedup over OS Paging on Galaxy S25+ and $9.2\times$ – $25.9\times$ on OnePlus 12.

6.3 Impact of Core Techniques

To understand the contribution of each core technique in mzCache, we conduct a detailed breakdown analysis of each technique. For focused comparison, we use Galaxy S25+ with Qwen3-0.6B, restoring from 50% remaining memory for a representative case unless noted.

Breakdown of performance improvements. Figure 10 shows the latency of allocation, reload, and prefill for Partial Offload and mzCache. For a clear analysis, we disable the overlapping feature of mzCache. mzCache achieves a $1.37\times$ speedup in allocation through parallelized memory allocation across CPU cores, $2.15\times$ in reload via zero-copy transfers and hybrid swap (further analyzed in Figure 12), and $2.33\times$ in prefill using a mobile-optimized attention kernel.

Benefits of overlapping. Figure 11a demonstrates the effect of overlapping by comparing four configurations: no overlap, allocation-reload overlap, reload-prefill overlap, and full overlap of all three operations. Overlapping allocation and reload achieves $1.10\times$ speedup, while overlapping reload and

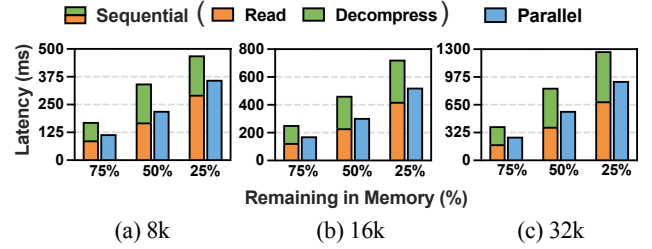


Figure 12: Sequential vs. parallel restoration time at different remaining memory levels. Sequential bars show stacked in-memory and storage restoration times.

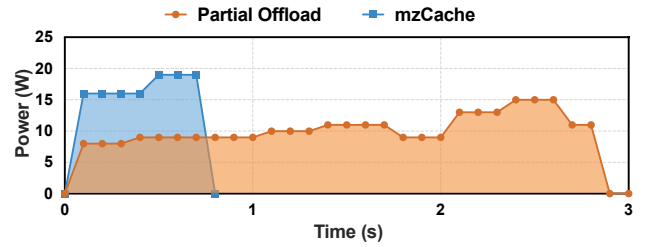


Figure 13: Power consumption during restoration and prefill. The shaded area under each curve indicates total energy consumption. Due to the reduced TTFT, mzCache consumes significantly less energy than Partial Offload despite its higher peak power.

prefill achieves larger gains of $1.41\times$. Full overlap of allocation, reload, and prefill yields $1.61\times$ speedup, demonstrating the cumulative benefit of overlapping all phases.

Impact of eviction ordering. Figure 11b evaluates the impact of our *backward-out* eviction policy. For comprehensive comparison, we implement two alternative eviction policies: *forward-out*, which evicts early layers first similar to LRU-based eviction, and *random-out*, averaged over 10 runs. All policies use forward-in restoration ordering. The results show that backward-out minimizes prefill stalls during inference, confirming the importance of evicting later layers first to preserve early layers needed for immediate computation.

Effect of hybrid swap. Figure 12 shows the effect of hybrid swap, demonstrating two key benefits: first, the speedup achieved by restoring via parallel paths compared to sequential restoration, and second, the effectiveness of load balancing between the two restore paths. The sequential case shows that the two restoration paths achieve similar latencies, confirming effective load balancing. As a result, parallel restoration significantly outperforms the sequential case.

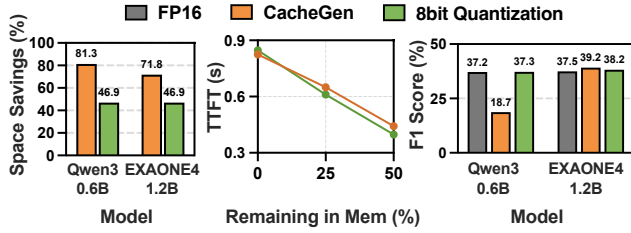


Figure 14: Comparison of 8-bit quantization and CacheGen: space savings, TTFT at different remaining memory levels, and F1 score relative to FP16.

6.4 Power and Energy Consumption

Figure 13 shows the power consumption trace during restoration and prefill on Galaxy S25+ with Qwen3-0.6B at 50% remaining memory and 32k context length. mzCache exhibits higher peak power at 19.2W compared to Partial Offload’s 14.6W, because mzCache fully utilizes storage, CPU, and GPU in parallel. Despite the higher peak power, mzCache completes the restoration and prefill process significantly faster, and as shown by the shaded area under each curve representing total energy consumption, mzCache consumes less energy overall. This demonstrates that mzCache’s latency reduction achieves better energy efficiency, an important consideration for battery-constrained mobile devices.

6.5 Compression Algorithm Comparison

mzCache treats KV cache compression as a modular component, allowing the algorithm to be replaced as needed. To analyze how the choice of compression algorithm affects performance, we compare 8-bit quantization [47] and CacheGen [36] on Galaxy S25+ in Figure 14. Space savings and TTFT are measured at 16k context length. At 50%, 8-bit quantization achieves lower TTFT due to faster decompression from in-memory swap. At 0%, CacheGen achieves better performance because its higher compression ratio reduces the size of compressed KV cache that must be loaded from storage. Both algorithms maintain comparable F1 scores on EXAONE-4.0-1.2B, while 8-bit quantization better preserves accuracy on Qwen3-0.6B, likely due to its larger KV hidden dimension and smaller model size. Notably, both algorithms are idempotent, so repeated compression and decompression do not accumulate error, keeping accuracy stable across swap cycles. These results show that the choice of compression algorithm should be tailored to the deployment scenario.

6.6 Verifying Real-world Deployment

To validate mzCache in a real mobile multitasking setting, we deploy an Android LLM application on Galaxy S25+ running Qwen3-0.6B with a 32k context length. To generate realistic

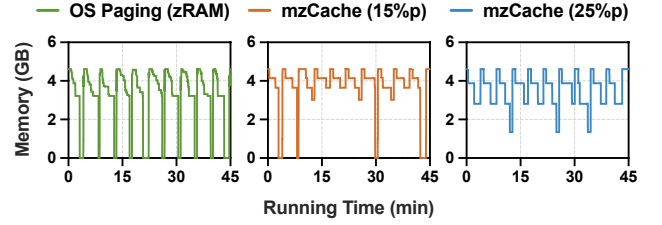


Figure 15: Total memory trace of an LLM application under real-world multitasking, comparing OS Paging (zRAM) with mzCache evicting approximately 15%p and 25%p of LLM memory under pressure. OS Paging is terminated by the LMK in every round, whereas mzCache (25%p) survives the entire session.

memory pressure, we interleave LLM use with everyday smartphone activity across ten rounds over ~45 minutes. Within each round, we interact with a range of popular consumer apps—social networking, video streaming, gaming, camera, and web browsing⁶—before triggering a prefill on the LLM application at the round’s end. We compare three configurations: OS Paging (zRAM), identical to the baseline in §6.1; mzCache (15%p), which evicts approximately 15%p of LLM memory upon each Android onTrimMemory callback; and mzCache (25%p), which evicts approximately 25%p.

Figure 15 shows the memory traces over the session. OS Paging is terminated by the LMK because the KV cache compresses poorly, forcing a cold start on every return. In contrast, mzCache (25%p) progressively evicts memory as pressure builds and survives all ten rounds, restoring context on each return without a cold start, showing that mzCache remains effective under real-world multitasking pressure where OS-level memory management fails. With a smaller eviction amount, however, mzCache (15%p) is terminated in four of the ten rounds. Since process termination is decided by the LMK and mzCache operates independently of the OS, it cannot prevent such terminations entirely. Deciding how much memory to evict and when becomes an important consideration for practical deployment, which we discuss in §7.

7 DISCUSSION

Extensibility across hardware. In our implementation, the CPU handles decompression while the GPU executes prefill, but these roles are not fixed and can be adapted to different hardware configurations; for instance, prefill can run on an NPU, or decompression can be offloaded to the GPU. More broadly, although we implement mzCache on a Qualcomm SoC, the underlying design applies to any SoC

⁶Instagram, Facebook, YouTube, TikTok, Netflix, Roblox, PUBG Mobile, Camera, SNOW, and Chrome.

with a unified memory architecture. Porting mzCache to another hardware platform thus requires replacing only the device-specific kernels to match the target SoC, while the rest of the system remains unchanged.

Pressure detection and eviction amount. mzCache decouples both memory pressure detection and the eviction amount from its core design, so each can be set per deployment. An application-level deployment can detect pressure via callbacks such as `onTrimMemory` and evict a larger amount to reduce the risk of LMK termination. A system-level deployment can instead use kernel signals such as `vmpressure` or `PSI`, and its low `oom_score_adj` places it far down the LMK's kill order, substantially reducing the risk that motivated the larger eviction amount [3]. The core mechanism is unchanged in either case, making mzCache deployable across the software stack, from third-party applications to system-level services.

Multi-context support. mzCache currently supports a single LLM context scenario. However, as LLM applications become more prevalent, multiple LLM applications may co-exist, each maintaining its own weights and KV caches. mzCache can be extended to handle such scenarios by adding coordination policies and data structures for managing multiple weights and KV caches. Such extensions are beyond the scope of this work and we remain them for future work.

Thermal throttling. To balance the two restore paths, mzCache relies on the sequential read and decompression throughputs obtained through offline profiling. Under sustained load, thermal throttling can shift these throughputs and thereby disrupt the balance between the two paths. One possible way to address this is to profile across multiple memory and CPU frequency combinations in advance, allowing mzCache to adapt to the current operating conditions at runtime. Such runtime adaptation, however, would require migrating already-evicted data between the two paths to match the new balance point.

Selectively activated models. Our design assumes that first-token generation touches all model weights and the entire KV cache, which holds for dense transformer models. Some architectures, however, activate only a subset at the first token: mixture-of-experts (MoE) models route each layer to a few experts [46], and sparse-attention models attend to only part of the KV cache [51]. For such models, the natural extension is to restore the activated subset first and defer the rest. Since which experts or KV chunks are active is input-dependent and not fully known before prefill, it must be predicted in advance—for example, from statistics of prior interactions. Incorporating such predictions into the restoration order would be an interesting direction for future work.

8 RELATED WORK

Offloading-based LLM inference. Several systems offload LLM data between GPU, CPU, or disk throughout inference to accommodate model weights and KV cache that exceed GPU memory [31, 47, 50]. Other work focuses on managing stored KV cache contexts across memory tiers to accelerate restoration upon reuse [12, 17, 25, 66, 67]. In all cases, the LLM system controls what to offload and when, with full knowledge of available memory. mzCache addresses a fundamentally different setting where eviction is externally imposed under unpredictable memory pressure, requiring a new system that adapts its memory management to unknown eviction amounts at runtime.

Mobile LLM inference systems. Existing optimizations for on-device LLM inference focus on accelerating runtime execution through heterogeneous co-execution across mobile processors [11, 21, 61], compiler and system stack optimizations for mobile SoCs [39, 53], and storage-based weight offloading for models exceeding device memory [59, 62, 64]. However, these are all runtime inference optimizations. In mobile multitasking environments, preserving the KV cache across interactions to reuse prior context creates a memory management challenge in the idle periods between inferences, a phase these systems do not manage. mzCache is the first system to address it.

Hybrid swap in mobile OS. In the mobile OS domain, some systems combine in-memory compressed swap with storage-backed swap to sustain responsiveness under memory pressure [30, 34, 35, 49], an idea that inspired our design. However, their general-purpose compression fails on the LLM KV cache (§3.1). Rather than modifying the OS, we adapt hybrid swap within the LLM inference system itself, specializing it for the KV cache and driving both paths in parallel with balanced load distribution, without OS kernel modifications.

9 CONCLUSION

In this paper, we present mzCache, an LLM inference system with specialized memory management for responsive inference in mobile multitasking environments. Our evaluation on commercial smartphones demonstrates that mzCache achieves 2.1–5.5× faster TTFT than partial offload and remains effective under realistic multitasking pressure. These results validate that dedicated memory management, rather than reliance on OS-based mechanisms, is essential for responsive LLM inference under multitasking.

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